Head & Neck Anatomy: Pharynx, Larynx, and Trachea

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1. The pharynx is a fibromuscular tube that acts as a common passage for air and food. Macroscopically, it extends from the base of the skull down to the lower border of which cartilage?
a. Cricoid cartilage (level with C6 vertebra)
b. Thyroid cartilage
c. Epiglottis
d. Arytenoid cartilage
e. Hyoid bone horns
2. The pharynx is divided into three distinct segments. The superior-most division situated directly behind the nasal cavities and above the soft palate is the:
a. Nasopharynx
b. Oropharynx
c. Laryngopharynx
d. Piriform recess
e. Fauces channel
3. Located within the mucosal roof and posterior wall of the nasopharynx is a prominent aggregation of lymphoid tissue. When pathologically enlarged in children, this mass is termed:
a. Palatine tonsil
b. Lingual tonsil
c. Tubal tonsil
d. Adenoids (pharyngeal tonsil)
e. Peyer's patch
4. The lateral wall of the nasopharynx features a high-yield clinical opening. This opening leads directly into an osteofibrous canal connecting to the middle ear cavity, known as the:
a. External acoustic meatus
b. Foramen rotundum
c. Pharyngeal recess (fossa of Rosenmüller)
d. Auditory (Eustachian) tube ostium
e. Incisive canal
5. Bounding the auditory tube opening posteriorly is a firm mucosal ridge produced by the underlying cartilaginous end of the tube, termed the:
a. Torus levatorius
b. Salpingopharyngeal fold
c. Passavant's ridge
d. Torus tubarius
e. Pterygoid hamulus
6. Situated immediately posterior to the torus tubarius is a deep, blind slit-like mucosal depression where nasopharyngeal carcinomas frequently originate. This depression is the:
a. Piriform recess
b. Oropharyngeal isthmus
c. Pharyngeal recess (fossa of Rosenmüller)

d. Vallecula
e. Glottis chamber

7. The oropharynx serves both respiratory and digestive paths. It lies posterior to the oral cavity proper, extending vertically from the soft palate down to the upper border of the:

a. Cricoid cartilage
b. Thyroid cartilage
c. Epiglottis
d. Hyoid bone body
e. Vocal folds

8. The lateral walls of the oropharynx house the palatine tonsils. Each tonsil is situated within a distinct fossa bounded anteriorly by the palatoglossal arch and posteriorly by the:

a. Salpingopharyngeal fold
b. Buccopharyngeal fascia
c. Superior pharyngeal constrictor
d. Palatopharyngeal arch
e. Pterygomandibular raphe

9. Bounded between the posterior base of the tongue and the anterior mucosal surface of the epiglottis are two small, shallow depressions where swallowed foreign bodies like fishbones can lodge. These are the:

a. Valleculae
b. Piriform recesses
c. Tonsillar crypts
d. Glottis slits
e. Meatus gutters

10. The inferior-most division of the pharynx is the laryngopharynx. It extends from the upper border of the epiglottis down to the commencement of the esophagus, lying posterior to the:

a. Nasal conchae
b. Hard palate
c. Soft palate
d. Larynx cavity
e. Tracheal bifurcation

11. Located on either side of the laryngeal inlet inside the laryngopharynx is a deep, gutter-like recess. This space is highly vulnerable to foreign body entrapment and is the:

a. Vallecula
b. Fossa of Rosenmüller
c. Pharyngeal bursa
d. Piriform recess
e. Fauces gate

12. Running immediately deep to the thin mucous membrane lining the piriform recess is a nerve that provides sensory control above the vocal folds. This nerve is the:

a. External laryngeal nerve
b. Recurrent laryngeal nerve
c. Glossopharyngeal nerve trunk

d. Internal laryngeal nerve (branch of CN X)
e. Hypoglossal nerve root

13. Histologically, because the nasopharynx is purely respiratory in function, its mucosal lining consists of:

a. Pseudostratified ciliated columnar epithelium with goblet cells
b. Stratified squamous non-keratinized epithelium
c. Keratinized stratified squamous epithelium
d. Simple squamous sheets
e. Simple cuboidal brush border epithelium

14. The oropharynx and laryngopharynx are subjected to mechanical friction during the passage of swallowed food boluses. Consequently, they are lined histologically by:

a. Simple columnar epithelium
b. Stratified squamous non-keratinized epithelium
c. Pseudostratified ciliated columnar epithelium
d. Transitional cells
e. Stratified cuboidal cells

15. The muscular wall of the pharynx consists of three outer circular constrictors. The muscle that takes structural origin from the pterygomandibular raphe, pterygoid hamulus, and mylohyoid line is the:

a. Superior pharyngeal constrictor
b. Middle pharyngeal constrictor
c. Inferior pharyngeal constrictor
d. Stylopharyngeus
e. Palatopharyngeus

16. The middle pharyngeal constrictor is a fan-shaped muscle layer that originates from the lower segment of the stylohyoid ligament and the:

a. Thyroid cartilage lamina
b. Cricoid cartilage arch
c. Horns (greater and lesser cornua) of the hyoid bone
d. Pterygomandibular raphe
e. Styloid process tip

17. The inferior pharyngeal constrictor is split into two distinct muscular segments. The lower segment, which arises from the cricoid cartilage and acts as a sphincter at the upper end of the esophagus, is the:

a. Thyropharyngeus muscle
b. Cricopharyngeus muscle
c. Stylopharyngeus
d. Salpingopharyngeus
e. Palatopharyngeus

18. Bounded between the upper thyropharyngeus fibers and lower cricopharyngeus fibers of the inferior constrictor is a posterior midline area of structural weakness known as:

a. Kiesselbach's triangle
b. Little's area
c. Woodburn's plexus
d. Killian's dehiscence
e. The piriform space

19. In older adults, uncoordinated contraction of the pharyngeal wall can cause the mucosa to herniate posteriorly through Killian's dehiscence, creating a diverticulum called:
a. Meckel's diverticulum
b. Zenker's diverticulum (pharyngeal pouch)
c. Rathke's cleft pouch
d. Branchial sinus
e. Thyroglossal cyst
20. The primary motor innervation to all the muscles of the pharynx (except stylopharyngeus) is mediated by fibers from the cranial accessory nerve traveling within the pharyngeal plexus of the:
a. Trigeminal nerve
b. Facial nerve
c. Vagus nerve (CN X)
d. Glossopharyngeal nerve
e. Hypoglossal nerve
21. The stylopharyngeus muscle is unique because it develops from the third pharyngeal arch. It receives its selective motor innervation from which cranial nerve?
a. Trigeminal nerve (V3 division)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Spinal accessory nerve (CN XI)
22. General somatic sensation from the mucous membrane of the nasopharynx is carried primarily by branches of the:
a. Ophthalmic nerve (V1)
b. Maxillary nerve (V2 via pharyngeal branches)
c. Mandibular nerve (V3)
d. Glossopharyngeal nerve
e. Vagus nerve
23. General sensation from the mucous membrane of the oropharynx, mediating the afferent limb of the gag reflex path, is carried by the:
a. Trigeminal nerve (V2)
b. Facial nerve (CN VII)
c. Glossopharyngeal nerve (CN IX)
d. Vagus nerve (CN X)
e. Hypoglossal nerve
24. General sensation from the mucous membrane of the laryngopharynx surrounding the laryngeal inlet is carried by the:
a. Glossopharyngeal nerve
b. Vagus nerve (CN X via the internal laryngeal nerve)
c. Mandibular nerve (V3)
d. Accessory nerve
e. Ansa cervicalis roots
25. Let's move to the larynx. The larynx is a specialized organ of voice production and a sphincteric air track. Macroscopically, its structural skeletal core is formed by how many total cartilages?
a. Three cartilages

b. Six cartilages
c. Nine cartilages (three unpaired, three paired)
d. Four cartilages
e. Twelve cartilages

26. Which of the following represents an unpaired skeletal cartilage component of the larynx?

a. Arytenoid cartilage
b. Thyroid cartilage
c. Cuneiform cartilage
d. Corniculate cartilage
e. Triticeal cartilage

27. The thyroid cartilage is the largest larynx cartilage. It consists of two flat lateral plates that fuse in the anterior midline to produce a prominent projection called the:

a. Laryngeal prominence (Adam's apple)
b. Superior cornu
c. Inferior cornu
d. Oblique line
e. Cricoid arch

28. The only larynx cartilage that forms a complete, rigid closed ring around the respiratory tract airway is the:

a. Thyroid cartilage
b. Cricoid cartilage
c. Epiglottis
d. Arytenoid cartilage
e. Tracheal ring

29. The cricoid cartilage is shaped like a signet ring. It consists of a narrow anterior arch and a broad posterior:

a. Cornu
b. Lamina
c. Styloid hook
d. Uncinate lip
e. Alar crest

30. The epiglottis is a leaf-shaped plate of cartilage that projects upward behind the hyoid bone. Its structural framework is composed of which resilient tissue?

a. Hyaline cartilage
b. White fibrocartilage
c. Yellow elastic cartilage
d. Dense compact bone
e. Calcified calcified sheets

31. The paired, pyramid-shaped larynx cartilages that sit on the upper borders of the cricoid lamina and articulate to regulate vocal fold positions are the:

a. Corniculate cartilages
b. Cuneiform cartilages
c. Arytenoid cartilages
d. Triticeal nodes
e. Sesamoidea plates

32. Each arytenoid cartilage features two prominent processes projecting from its base. These are the:

a. Superior and inferior cornua
b. Anterior vocal process and lateral muscular process
c. Uncinate process and pterygoid process
d. Mastoid and styloid tips
e. Sternal and costal crests

33. The cricothyroid joint is a synovial joint between the inferior horn of the thyroid cartilage and the cricoid cartilage. Rotational movement at this joint results in:
a. Complete closure of the epiglottis
b. Tensing or lengthening of the vocal folds
c. Abduction of the vocal processes
d. Elevation of the hyoid bone
e. Relaxation of the thyropharyngeus muscle

34. The internal cavity of the larynx is divided into regions by two pairs of horizontal mucosal folds. The upper, vascularized pair that does not produce voice is the:
a. True vocal folds
b. Vestibular folds (false vocal cords)
c. Vocal ligaments
d. Aryepiglottic folds
e. Glottis margins

35. The lower, sharp-edged mucosal folds that lack a submucosa, look white clinically, and contain the vocal ligament and vocalis muscle are the:
a. Vestibular folds
b. True vocal folds (vocal cords)
c. Palatoglossal arches
d. Salpingopharyngeal folds
e. Ventricle lips

36. Bounded between the upper vestibular folds and the lower true vocal folds on either side of the larynx is a deep lateral spindle-shaped recess called the:
a. Vestibule of the larynx
b. Saccule of the larynx
c. Ventricle (sinus) of the larynx
d. Rima glottidis
e. Piriform recess

37. The narrow horizontal fissure or space bounded between the two true vocal folds on either side is the:
a. Rima vestibuli
b. Rima glottidis
c. Conus elasticus
d. Inlet of larynx
e. Pharyngeal recess

38. Muscles of the larynx alter vocal cord positions. The only intrinsic muscle of the larynx that abducts the vocal folds (opens the rima glottidis wide) is the:
a. Lateral cricoarytenoid
b. Posterior cricoarytenoid muscle ("safety muscle of larynx")
c. Cricothyroid muscle
d. Thyroarytenoid
e. Interarytenoid

39. Bilateral paralysis of the posterior cricoarytenoid muscles due to iatrogenic nerve injury causes a catastrophic clinical presentation characterized by:
a. Inability to swallow fluids
b. Immediate, total closure of the rima glottidis leading to asphyxiation
c. Complete loss of taste on the tongue
d. Soft tissue swelling of the neck skin
e. Profuse secretomotor dry mouth
40. The intrinsic muscle of the larynx that acts as the primary adductor of the vocal folds, closing the rima glottidis, is the:
a. Posterior cricoarytenoid
b. Thyroarytenoid
c. Lateral cricoarytenoid muscle (along with transverse arytenoid)
d. Cricothyroid
e. Vocalis muscle
41. The cricothyroid muscle is an intrinsic laryngeal muscle situated on the external face of the larynx. When it contracts, it pulls the thyroid arch downward and forward, acting to:
a. Abduct the vocal cords
b. Tense or elongate the vocal folds (raising vocal pitch)
c. Relax the vocal ligament
d. Close the laryngeal inlet completely
e. Elevate the epiglottis flap
42. All intrinsic muscles of the larynx develop from the sixth pharyngeal arch mesenchyme, with one exception that develops from the fourth arch. This exception is the:
a. Posterior cricoarytenoid muscle
b. Lateral cricoarytenoid muscle
c. Cricothyroid muscle
d. Thyroarytenoid muscle
e. Vocalis muscle
43. Because of its distinct pharyngeal arch origin, the cricothyroid muscle is supplied motorically by a branch of the superior laryngeal nerve from CN X, known as the:
a. Internal laryngeal nerve
b. External laryngeal nerve
c. Recurrent laryngeal nerve
d. Pharyngeal ramus
e. Cardiac branch
44. All other intrinsic muscles of the larynx (except cricothyroid) receive their motor innervation from which nerve channel?
a. Internal laryngeal nerve
b. External laryngeal nerve
c. Recurrent laryngeal nerve (branch of CN X)
d. Glossopharyngeal nerve
e. Hypoglossal nerve
45. Sensory innervation to the mucous membrane of the larynx *superior* to the level of the true vocal folds is mediated entirely by the:
a. External laryngeal nerve
b. Internal laryngeal nerve (branch of the superior laryngeal nerve from CN X)

c. Recurrent laryngeal nerve
d. Glossopharyngeal nerve
e. Lingual nerve

46. Accidental injury or topical blocking of the internal laryngeal nerve results in anesthesia of the upper larynx mucosa. This causes a loss of which vital protective reflex?

a. Corneal blink reflex
b. Gag reflex
c. Cough reflex (allowing swallowed food to enter the airway silently)
d. Pupillary constriction reflex
e. Accommodation reflex

47. Sensory innervation to the mucous membrane of the larynx *inferior* to the level of the true vocal folds is carried by the:

a. Internal laryngeal nerve
b. External laryngeal nerve
c. Recurrent laryngeal nerve
d. Descendens cervicalis
e. Superior laryngeal trunk

48. Histologically, the vast majority of the larynx cavity is lined by respiratory epithelium. However, regions subjected to mechanical friction—such as the epiglottis tip and the true vocal folds—are lined by:

a. Pseudostratified ciliated columnar epithelium
b. Simple columnar epithelium
c. Stratified squamous non-keratinized epithelium
d. Transitional epithelium
e. Simple cuboidal epithelium

49. Microscopic evaluation of a chronic vocal cord nodule ("singer's node") reveals focal thickening and hyperkeratosis involving the stratified squamous epithelium of the:

a. Vestibular folds
b. True vocal folds
c. Epiglottis base
d. Aryepiglottic fold
e. Tracheal bifurcation

50. Let's move to the trachea. The trachea (windpipe) is a mobile cartilaginous tube that extends down from the lower border of the cricoid cartilage. At which thoracic level does it normally bifurcate into the right and left primary bronchi?

a. Upper border of T2 vertebra
b. Lower border of T4 or T5 vertebra (sternal angle level)
c. Lower border of T8 vertebra
d. Level of T12 vertebra
e. Level of C7 vertebra

51. Macroscopically, the patency of the trachea lumen is maintained by a series of how many C-shaped, deficient-posteriorly cartilage rings?

a. 5 to 8 rings
b. 16 to 20 rings
c. 30 to 40 rings
d. 12 rings exactly

e. 50 rings

52. The structural cartilage rings that form the framework of the trachea are composed histologically of which cartilage variety?

a. Yellow elastic cartilage
b. White fibrocartilage
c. Hyaline cartilage
d. Spongy trabecular bone
e. Calcified calcified sheets

53. The posterior deficiency of the C-shaped tracheal cartilage rings is closed by a thick band of smooth muscle tissue and elastic fibers, known as the:

a. Cricothyroid muscle
b. Vocalis muscle
c. Trachealis muscle
d. Inferior pharyngeal constrictor
e. Esophageal circular layer

54. The trachealis muscle sits directly anterior to the esophagus. When a large food bolus passes down the esophagus during swallowing, the trachealis muscle relaxes to allow:

a. The trachea to shorten vertically
b. Posterior bulging of the esophagus into the trachea lumen without structural blockage
c. Sudden high-pressure air flows
d. Total closure of the glottis
e. The thyroid gland to expand

55. Histologically, the internal surface of the trachea is lined by a classic respiratory mucosa. The surface epithelium is microscopically characterized as:

a. Simple squamous epithelium
b. Stratified squamous non-keratinized epithelium
c. Pseudostratified ciliated columnar epithelium with abundant goblet cells
d. Transitional epithelium
e. Simple cuboidal brush border epithelium

56. Under the microscope, the cilia of the tracheal respiratory epithelial cells beat continuously in a coordinated fashion. Their primary physiological direction is targeted:

a. Downward toward the lungs
b. Upward toward the larynx and pharynx to clear mucus and trapped dust particles
c. Laterally toward the thyroid lobes
d. Backward into the esophagus wall
e. In a chaotic random circle

57. The thick connective tissue submucosa of the trachea contains an abundance of mixed exocrine glands that secrete a fluid film onto the epithelial surface. These are the:

a. Bowman's glands
b. Tracheal (seromucous) glands
c. Sebaceous glands
d. Von Ebner glands
e. Meibomian glands

58. High-pressure arterial blood is supplied to the cervical portion of the trachea primarily via branches derived from the:

a. Superior thyroid artery
b. Inferior thyroid artery
c. Internal thoracic artery
d. Maxillary artery
e. Ascending pharyngeal artery

59. Venous blood from the cervical trachea drains backward into the extensive venous network of the lower neck, emptying into the:

a. Internal jugular vein
b. External jugular vein
c. Inferior thyroid veins (or brachiocephalic veins)
d. Subclavian vein
e. Anterior jugular venous arch

60. The sensory nerve fibers that supply pain and touch awareness to the mucous membrane of the trachea travel within the:

a. Internal laryngeal nerve
b. External laryngeal nerve
c. Recurrent laryngeal nerve (branch of CN X)
d. Phrenic nerve
e. Ansa cervicalis loops

61. A vital ridge or internal shelf of cartilage is situated at the absolute bifurcation point of the trachea into the primary bronchi. This highly sensitive landmark is the:

a. Cricoid arch
b. Pyramidal eminence
c. Carina
d. Promontory
e. Epiglottis cushion

62. Foreign bodies that bypass the larynx into the trachea frequently land at the carina. Irritation of the sensory nerve endings at the carina triggers a powerful, explosive:

a. Gag reflex
b. Cough reflex
c. Deglutition cycle
d. Sneeze reflex
e. Pupillary dilatation

63. When an inhaled foreign object passes past the carina into the bronchial tree, it characteristically enters the right primary bronchus rather than the left because the right bronchus is:

a. Longer and more horizontal
b. Shorter, wider, and more vertical (in direct line with the trachea)
c. Completely closed by a mucosal fold
d. Completely missing any cartilage rings
e. Situated superior to the larynx plane

64. An emergency surgical procedure to establish an airway when the upper respiratory tract is completely blocked is a tracheostomy. The cervical trachea is accessed safely along the anterior midline of the neck by dividing the:

a. Pterygomandibular raphe
b. Linea alba of the neck (midline cervical raphe)
c. Platysma sheet horizontally

d. Masseter muscle sling
e. Stylohyoid ligament path

65. During a standard tracheostomy, the surgical incision passes between the infrahyoid strap muscles. Which endocrine structure crossing the anterior surface of the 2nd, 3rd, and 4th tracheal rings must be divided or retracted?

a. Parathyoids
b. Thymus
c. Thyroid gland isthmus
d. Submandibular gland deep lobe
e. Parotid duct

66. Running vertically along the right and left borders of the trachea, in close proximity to the inferior thyroid artery, are the recurrent laryngeal nerves. A surgeon operating near the lateral walls of the trachea must protect these nerves to prevent:

a. Total numbness of the tongue tip
b. Dry eyes due to lacrimal shutdown
c. Vocal fold paralysis and hoarseness
d. Sudden hyperparathyroidism
e. Loss of taste on the palate

67. Microscopic examination of the basement membrane of the tracheal respiratory epithelium reveals a remarkably thick layer composed of tightly packed:

a. Keratin scales
b. Collagen fibers and extracellular matrix
c. Calcified bone chips
d. Odontoblast processes
e. Striated muscle bundles

68. The outermost connective tissue layer that wraps around the external surface of the tracheal cartilage rings and anchors the trachea to the surrounding structures in the neck is the:

a. Mucosa
b. Submucosa
c. Lamina propria
d. Adventitia
e. Perichondrium sleeve

69. Pathological compression of the trachea can occur due to a massive, abnormal enlargement of the thyroid gland, a clinical condition known as a:

a. Ranula
b. Goiter
c. Branchial cyst
d. Mucocele
e. Dental abscess

70. The small lumen or cleft situated between the vestibular folds and the vocal folds within the larynx chamber is the ventricle. A superior outpocketing of this ventricle lined with mucous glands that lubricate the vocal cords is the:

a. Rima glottidis
b. Conus elasticus
c. Laryngeal saccule
d. Piriform sinus
e. Vallecula cup